INVERSE TRIGONOMETRY LEVEL 3

ADVANCED MCQs – JEE MAIN STYLE (~130 QUESTIONS)

SECTION 1: NUMERICAL–CONCEPTUAL BASICS (Q1–Q35)

1. If $\sin^{-1} x + \cos^{-1} x = \theta$, then θ equals

   A. 0

   B. $\frac{\pi}{4}$

   C. $\frac{\pi}{2}$

   D. π

2. If $\tan^{-1} x + \cot^{-1} x = \phi$ for $x > 0$, then ϕ equals

   A. 0

   B. $\frac{\pi}{4}$

   C. $\frac{\pi}{2}$

   D. π

3. If $\sin^{-1} x = \cos^{-1} y$ with $x, y \in [0,1]$, then

   A. $x = y$

   B. $x^2 + y^2 = 1$

   C. $x + y = 1$

   D. $x - y = 1$

4. The value of $\sin^{-1} x + \sin^{-1} \sqrt{1 - x^2}$ for $x \in [0,1]$ is

   A. 0

   B. $\frac{\pi}{2}$

   C. π

   D. Depends on x

5. If $\alpha = \sin^{-1} \frac{3}{5}$ and $\beta = \cos^{-1} \frac{4}{5}$, then $\alpha + \beta$ equals

   A. 0

   B. $\frac{\pi}{4}$

   C. $\frac{\pi}{2}$

   D. π

6. If $\theta = \sin^{-1}\frac{12}{13}$, then $\tan\ \theta$ equals

A. $\frac{5}{12}$

B. $\frac{12}{5}$

C. $\frac{13}{5}$

D. $\frac{5}{13}$

7. If $\phi = \tan^{-1}\frac{3}{4}$, then $\sin\ \phi$ equals

A. $\frac{3}{5}$

B. $\frac{4}{5}$

C. $\frac{5}{3}$

D. $\frac{5}{4}$

8. The value of $\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3}$ is

A. $\frac{\pi}{6}$

B. $\frac{\pi}{4}$

C. $\frac{\pi}{3}$

D. $\frac{\pi}{2}$

9. The value of $\tan^{-1} 2 + \tan^{-1} 3$ is

A. $\frac{\pi}{4}$

B. $\frac{\pi}{2}$

C. π

D. 0

10. If $\theta = \tan^{-1} x$, then $\sin\ 2\theta$ equals

A. $\frac{2x}{1+x^2}$

B. $\frac{1-x^2}{1+x^2}$

C. $\frac{2x}{1-x^2}$

D. $\frac{1+x^2}{1-x^2}$

11. If $\theta = \tan^{-1} x$, then $\cos 2\theta$ equals

A. $\frac{1-x^2}{1+x^2}$

B. $\frac{2x}{1+x^2}$

C. $\frac{1+x^2}{1-x^2}$

D. $\frac{2}{1+x^2}$

12. If $\theta = \sin^{-1} x$, then $\cos 2\theta$ equals

A. $1 - 2x^2$

B. $2x^2 - 1$

C. $1 - x^2$

D. $x^2 - 1$

13. If $0 < x < 1$, then $\tan^{-1}\left(\frac{2x}{1-x^2}\right)$ equals

A. $\tan^{-1} x$

B. $2\tan^{-1} x$

C. $\frac{\pi}{2} - 2\tan^{-1} x$

D. $\frac{\pi}{2}$

14. The value of $\tan^{-1}\frac{1}{x} + \tan^{-1} x$ for $x > 0$ is

A. 0

B. $\frac{\pi}{4}$

C. $\frac{\pi}{2}$

D. π

15. If $\sin^{-1} x = \tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)$, then $|x|$ is

A. ≥ 1

B. ≤ 1

C. > 1

D. < 0

16. The value of $\sin^{-1} x + \cos^{-1} x$ is

A. 0

B. $\frac{\pi}{2}$

C. π

D. Depends on x

17. If $\alpha = \sin^{-1} \frac{5}{13}$, then $\sec \ \alpha$ equals

A. $\frac{12}{5}$

B. $\frac{13}{5}$

C. $\frac{13}{12}$

D. $\frac{5}{12}$

18. If $\beta = \cos^{-1} \frac{5}{13}$, then $\csc \ \beta$ equals

A. $\frac{12}{5}$

B. $\frac{13}{12}$

C. $\frac{13}{5}$

D. $\frac{5}{12}$

19. The value of $\tan^{-1} x + \tan^{-1} \frac{1-x}{1+x}$ is

A. 0

B. $\frac{\pi}{4}$

C. $\frac{\pi}{2}$

D. π

20. If $\sin^{-1} x = \cos^{-1} x$, then x equals

A. 0

B. $\frac{1}{2}$

C. $\frac{1}{\sqrt{2}}$

D. $-\frac{1}{\sqrt{2}}$

21. If $\tan^{-1} x + \tan^{-1} y = \frac{\pi}{4}$, then $\frac{x+y}{1-xy}$ equals

A. 0

B. 1

C. -1

D. 2

22. If $\tan^{-1} a + \tan^{-1} b + \tan^{-1} c = \pi$, then

A. $a + b + c = abc$

B. $ab + bc + ca = 0$

C. $a + b + c = 0$

D. $abc = 1$

23. The value of $\sin^{-1}\left(\frac{2t}{1+t^2}\right)$ in terms of $\tan^{-1} t$ is

A. $\tan^{-1} t$

B. $2\tan^{-1} t$ (with suitable conditions)

C. $\frac{\pi}{2} - 2\tan^{-1} t$

D. $\pi - 2\tan^{-1} t$

24. The value of $\cos^{-1}\left(\frac{1-t^2}{1+t^2}\right)$ in terms of $\tan^{-1} t$ is

A. $2\tan^{-1} t$

B. $\tan^{-1} t$

C. $\frac{\pi}{2} - 2\tan^{-1} t$

D. $\sin^{-1} t$

25. If $x = \sin^{-1} \frac{3}{5}$ and $y = \cos^{-1} \frac{4}{5}$, then $\sin(x + y)$ equals

A. 0

B. 1

C. $\frac{7}{25}$

D. $\frac{24}{25}$

26. If $\alpha = \sin^{-1} \frac{4}{5}$ and $\beta = \cos^{-1} \frac{3}{5}$, then $\cos(\alpha + \beta)$ equals

A. 0

B. 1

C. -1

D. $\frac{7}{25}$

27. The expression $\tan^{-1} \sqrt{3} + \tan^{-1} \frac{1}{2}$ equals

A. $\frac{\pi}{6}$

B. $\frac{\pi}{4}$

C. $\frac{\pi}{3}$

D. $\frac{\pi}{2}$

28. If $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \pi$, then one relation between x, y, z is

A. $x + y + z = xyz$

B. $xy + yz + zx = 1$

C. $x + y + z = 0$

D. $xyz = 0$

29. The equation $\tan^{-1} x = \sin^{-1} \frac{2x}{1+x^2}$ holds for

A. All real x

B. $|x| < 1$

C. $|x| > 1$

D. $x \geq 0$ only

30. If $\alpha = \cos^{-1} \frac{4}{5}$, then $\tan \alpha$ equals

A. $\frac{3}{4}$

B. $\frac{4}{3}$

C. $\frac{5}{4}$

D. $\frac{5}{3}$

31. If $\theta = \tan^{-1} x$, then $\sin \theta + \cos \theta$ equals

A. $\frac{x+1}{\sqrt{1+x^2}}$

B. $\frac{x+1}{1+x^2}$

C. $\sqrt{1 + x^2}$

D. $\frac{1-x}{\sqrt{1+x^2}}$

32. If $\theta = \tan^{-1} x$, then $\sin \theta - \cos \theta$ equals

A. $\frac{x-1}{\sqrt{1+x^2}}$

B. $\frac{1-x}{\sqrt{1+x^2}}$

C. $\frac{1+x}{\sqrt{1+x^2}}$

D. $\frac{x+1}{1+x^2}$

33. If $\alpha = \sin^{-1} x$ and $\beta = \cos^{-1} x$, then $\sin(\alpha + \beta)$ equals

A. 0

B. 1

C. x

D. $\sqrt{1 - x^2}$

34. If $\alpha = \sin^{-1} x$ and $\beta = \cos^{-1} x$, then $\cos(\alpha + \beta)$ equals

A. 0

B. 1

C. -1

D. $2x^2 - 1$

35. If $\theta = \sin^{-1} x$ and $|x| \leq 1$, then $\sin\theta \cos\theta$ equals

A. $x\sqrt{1 - x^2}$

B. $1 - x^2$

C. x^2

D. $\sqrt{1 - x^2}$

SECTION 2: DOMAIN, RANGE & GRAPH-BASED MCQs (Q36–Q70)

36. The domain of $f(x) = \sin^{-1}(2x - 1)$ is

A. $(-\infty, \infty)$

B. $[0,1]$

C. $[0,1]$ shifted so that $2x - 1 \in [-1,1]$

D. $[0,2]$

37. The domain of $f(x) = \cos^{-1}(2x^2 - 1)$ is

A. All real x

B. $|x| \leq 1$

C. $|x| \geq 1$

D. $[-1,1]$ only

38. The domain of $f(x) = \tan^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ is

A. All real x

B. $|x| \neq 1$

C. $|x| < 1$

D. $|x| > 1$

39. The range of $y = \sin^{-1}(2x - 1)$ is

A. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

B. $[0, \pi]$

C. $[-\pi, \pi]$

D. Depends on x

40. The range of $y = \cos^{-1}(2x - 1)$ is

A. $[0, \pi]$

B. $[-\pi, \pi]$

C. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

D. $[0,2\pi]$

41. The range of $y = \tan^{-1}(2x)$ is

A. $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

B. $[0, \pi]$

C. $(-\pi, \pi)$

D. $[0,2\pi]$

42. The range of $y = \sin^{-1}(\sin\ x)$ (principal) is

A. $[0, \pi]$

B. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

C. $[\pi, 2\pi]$

D. $(-\pi, \pi)$

43. The range of $y = \cos^{-1}(\cos\ x)$ is

A. $[0, \pi]$

B. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

C. $[-\pi, \pi]$

D. $[0,2\pi]$

44. The function $f(x) = \sin^{-1}\ x$ is

A. Decreasing on $[-1,1]$

B. Increasing on $[-1,1]$

C. Constant on $[-1,1]$

D. Decreasing then increasing

45. The function $g(x) = \cos^{-1}\ x$ is

A. Increasing on $[-1,1]$

B. Decreasing on $[-1,1]$

C. Constant on $[-1,1]$

D. Increasing then decreasing

46. The function $h(x) = \tan^{-1}\ x$ is

A. Increasing on $\mathbb{R}$

B. Decreasing on $\mathbb{R}$

C. Constant

D. Neither increasing nor decreasing

47. The minimum value of $\sin^{-1}(x) + \cos^{-1}(x)$ is

A. 0

B. $\frac{\pi}{4}$

C. $\frac{\pi}{2}$

D. π

48. The maximum value of $\sin^{-1}(x) + \cos^{-1}(x)$ is

A. $\frac{\pi}{2}$

B. π

C. 0

D. 2π

49. The range of the function $f(x) = \sin^{-1}\ x + \cos^{-1}\ x$ is

A. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

B. $\left[\frac{\pi}{2}, \pi\right]$

C. $\{\frac{\pi}{2}\}$

D. $[0, \pi]$

50. The range of $f(x) = \tan^{-1}\ x + \tan^{-1}\ \frac{1}{x}$ for $x > 0$ is

A. $\{\frac{\pi}{2}\}$

B. $\{\pi\}$

C. $(0, \pi)$

D. $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

51. The graph of $y = \sin^{-1}\ x$ is obtained by reflecting the graph of

A. $y = \sin\ x$ in the line $y = x$ with restricted domain

B. $y = \cos\ x$ in x-axis

C. $y = \tan\ x$ in y-axis

D. $y = \sin\ x$ in x-axis

52. The graph of $y = \cos^{-1}\ x$ is obtained from $y = \cos\ x$ by

A. translation only

B. reflection in the line $y = x$ with restriction

C. reflection in x-axis

D. rotation

53. The function $f(x) = \sin^{-1}\ (x^2)$ has domain

A. $\mathbb{R}$

B. $[-1,1]$

C. $x: x^2 \leq 1$

D. Both B and C

54. The function $g(x) = \cos^{-1}\ (2x^2 - 3)$ has domain given by

A. $|2x^2 - 3| \leq 1$

B. $|2x^2 - 3| \geq 1$

C. $2x^2 - 3 \geq 1$

D. $2x^2 - 3 \leq -1$

55. The function $h(x) = \tan^{-1}\ (x^2)$ has domain

A. $\mathbb{R}$

B. $[-1,1]$

C. $[0, \infty)$

D. $(-\infty, 0]$

56. The inequality $\sin^{-1}\ x > 0$ holds for

A. $x > 0$

B. $x \geq 0$

C. $x \in (0,1]$

D. $x \in [-1,0]$

57. The inequality $\cos^{-1} x < \frac{\pi}{2}$ holds for

A. $x > 0$

B. $x < 0$

C. all $x \in [-1,1]$

D. no x

58. The inequality $\tan^{-1} x \geq 0$ holds for

A. $x \geq 0$

B. $x > 0$ only

C. $x \leq 0$

D. all x

59. The least value of $\cos^{-1} x$ is

A. 0

B. $-\frac{\pi}{2}$

C. $\frac{\pi}{2}$

D. π

60. The greatest value of $\sin^{-1} x$ is

A. 0

B. $\frac{\pi}{2}$

C. π

D. 1

61. The range of $f(x) = \tan^{-1}(\sin x)$ is

A. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

B. $[-\tan^{-1} 1, \tan^{-1} 1]$

C. $[-1,1]$

D. $[0, \pi]$

62. The set of all values taken by $\sin^{-1}(x) + \sin^{-1}(1-x)$ when $x \in [0,1]$ is

A. a single point $\frac{\pi}{2}$

B. $[0, \pi]$

C. $[0, \frac{\pi}{2}]$

D. $[\frac{\pi}{2}, \pi]$

63. The function $f(x) = \sin^{-1}\ x$ is

A. onto from $[-1,1] \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

B. not one-one

C. many-one from $[-1,1]$ to same range

D. both A and C

64. The function $g(x) = \cos^{-1}\ x$ is

A. one-one decreasing on $[-1,1]$

B. one-one increasing

C. not invertible

D. constant

65. The function $h(x) = \tan^{-1}\ x$ maps $\mathbb{R}$

A. onto $(-\pi, \pi)$

B. onto $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

C. into $[0, \pi]$

D. into $[0,2\pi]$

66. The inverse of $y = \sin^{-1}\ x$ is

A. $y = \sin\ x, x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

B. $y = \cos\ x$ on $[0, \pi]$

C. $y = \tan\ x$ on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

D. none

67. The inverse of $y = \tan^{-1}\ x$ is

A. $y = \tan\ x$ on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

B. $y = \tan\ x$ on $(0, \pi)$

C. $y = \cot\ x$

D. none

68. The graph of $y = \tan^{-1}\ x$ has horizontal asymptotes at

A. $y = 0$

B. $y = \pm\frac{\pi}{2}$

C. $y = \pm\pi$

D. none

69. As $x \to \infty$, $\tan^{-1} x \to$

A. 0

B. $\frac{\pi}{2}$

C. $-\frac{\pi}{2}$

D. π

70. As $x \to -\infty$, $\tan^{-1} x \to$

A. 0

B. $\frac{\pi}{2}$

C. $-\frac{\pi}{2}$

D. $-\pi$

SECTION 3: EQUATIONS, INEQUALITIES & MIXED (Q71–Q110)

71. The solution of $\sin^{-1} x = \frac{\pi}{3}$ in $[-1,1]$ is

A. $x = \frac{1}{2}$

B. $x = \frac{\sqrt{3}}{2}$

C. $x = \frac{\sqrt{2}}{2}$

D. no solution

72. The solution of $\cos^{-1} x = \frac{\pi}{3}$ in $[-1,1]$ is

A. $x = \frac{1}{2}$

B. $x = \frac{\sqrt{3}}{2}$

C. $x = -\frac{1}{2}$

D. no solution

73. The solution of $\tan^{-1} x = \frac{\pi}{3}$ in $\mathbb{R}$ is

A. none

B. exactly one principal solution $x = \sqrt{3}$

C. infinitely many

D. two solutions

74. The equation $\sin^{-1} x + \sin^{-1} x = \frac{\pi}{2}$ gives

A. $x = 0$

B. $x = \frac{\sqrt{2}}{2}$

C. $x = 1$

D. $x = -\frac{\sqrt{2}}{2}$

75. The equation $\cos^{-1} x + \cos^{-1} x = \pi$ gives

A. $x = 0$

B. $x = \frac{1}{2}$

C. $x = -\frac{1}{2}$

D. $x = 1$

76. The domain of $f(x) = \sin^{-1}(3x - 2)$ is

A. $\left[\frac{1}{3}, 1\right]$

B. $[0,1]$

C. $\left[\frac{1}{3}, \frac{5}{3}\right]$

D. $(-\infty, \infty)$

77. The inequality $\sin^{-1} x \leq \frac{\pi}{4}$ implies

A. $x \leq \frac{\sqrt{2}}{2}$

B. $x < \frac{\sqrt{2}}{2}$

C. $x \geq \frac{\sqrt{2}}{2}$

D. $x \leq 0$

78. The inequality $\cos^{-1} x \geq \frac{\pi}{3}$ implies

A. $x \leq \frac{1}{2}$

B. $x < \frac{1}{2}$

C. $x \geq \frac{1}{2}$

D. $x \geq 0$

79. The inequality $\tan^{-1} x < \frac{\pi}{6}$ implies

A. $x < \frac{1}{\sqrt{3}}$

B. $x > \frac{1}{\sqrt{3}}$

C. $x = \frac{1}{\sqrt{3}}$

D. no solution

80. The number of solutions of $\sin^{-1} x = \cos^{-1} x$ in $[-1,1]$ is

A. 0

B. 1

C. 2

D. infinite

81. The number of solutions of $\sin^{-1} x = \tan^{-1} x$ in $[-1,1]$ is

A. 0

B. 1

C. 2

D. 3

82. The number of solutions of $\cos^{-1} x = \tan^{-1} x$ in $[-1,1]$ is

A. 0

B. 1

C. 2

D. infinite

83. The solution set of $\sin^{-1} x + \sin^{-1}(1 - x) = \frac{\pi}{2}$ in $[0,1]$ is

A. entire interval $[0,1]$

B. only $x = \frac{1}{2}$

C. only $x = 0$ or 1

D. no solution

84. The function $f(x) = \sin^{-1} x$ is concave on

A. $[-1,1]$

B. $[-1,0]$ and convex on $[0,1]$

C. convex on $[-1,0]$ and concave on $[0,1]$

D. none

85. The function $g(x) = \cos^{-1} x$ is

A. concave on $[-1,1]$

B. convex on $[-1,1]$

C. linear on $[-1,1]$

D. none

86. The limit $\lim_{x \to 0} \frac{\sin^{-1} x}{x}$ equals

A. 0

B. 1

C. $\frac{\pi}{2}$

D. ∞

87. The limit $\lim_{x \to 0} \frac{\tan^{-1} x}{x}$ equals

A. 0

B. 1

C. $\frac{\pi}{2}$

D. ∞

88. The derivative of $\sin^{-1} x$ at $x = 0$ is

A. 0

B. 1

C. $\frac{1}{\sqrt{1-0^2}} = 1$ so 1

D. ∞

89. The derivative of $\cos^{-1} x$ at $x = 0$ is

A. 0

B. 1

C. -1

D. undefined

90. The derivative of $\tan^{-1} x$ at $x = 1$ is

A. 0

B. $\frac{1}{2}$

C. $\frac{1}{1+1^2} = \frac{1}{2}$

D. 1

91. The function $y = \sin^{-1}(2x\sqrt{1-x^2})$ can be expressed as

A. $\sin^{-1}(\sin\ 2x)$

B. $2x$

C. $2\sin^{-1}\ x$

D. $\pi - 2x$

92. If $y = \cos^{-1}(1-2x^2)$, then y equals

A. $2\sin^{-1}\ x$ (for $|x| \leq 1$)

B. $\sin^{-1}\ x$

C. $2\tan^{-1}\ x$

D. $\pi - 2\sin^{-1}\ x$

93. If $\alpha = \sin^{-1}\ x$, then $\tan\ \alpha$ equals

A. $\frac{x}{\sqrt{1-x^2}}$

B. $\frac{\sqrt{1-x^2}}{x}$

C. $x\sqrt{1-x^2}$

D. none

94. If $\beta = \cos^{-1}\ x$, then $\tan\ \beta$ equals

A. $\frac{\sqrt{1-x^2}}{x}$

B. $\frac{x}{\sqrt{1-x^2}}$

C. $x\sqrt{1-x^2}$

D. none

95. The expression $\tan^{-1}\ x + \tan^{-1}\frac{1-x}{1+x}$ is independent of

A. x

B. $\tan^{-1}\ x$

C. the domain of x

D. all of these

96. The composition $\tan(\cos^{-1}\ x)$ equals

A. $\frac{\sqrt{1-x^2}}{x}$

B. $\frac{x}{\sqrt{1-x^2}}$

C. $x\sqrt{1-x^2}$

D. none

97. The composition $\cot(\sin^{-1} x)$ equals

A. $\frac{\sqrt{1-x^2}}{x}$

B. \dfrac{x}{\sqrt{1-x^{2}}}

C. $x\sqrt{1-x^2}$

D. none

98. The expression $\sin^{-1} x - \cos^{-1} x$ equals

A. 0

B. $-\frac{\pi}{2}$

C. $\frac{\pi}{2}$

D. depends on x

99. The expression $\tan^{-1} x - \tan^{-1} \frac{1}{x}$ for $x > 0$ equals

A. 0

B. $-\frac{\pi}{2}$

C. $\frac{\pi}{2}$

D. π

100. The range of $f(x) = \sin^{-1} |x|$ for $x \in [-1,1]$ is

A. $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

B. $[0, \frac{\pi}{2}]$

C. $[\frac{\pi}{2}, \pi]$

D. $[-\pi, 0]$

101. The range of $f(x) = \cos^{-1} |x|$ for $x \in [-1,1]$ is

A. $[0, \frac{\pi}{2}]$

B. $[\frac{\pi}{2}, \pi]$

C. $[0, \pi]$

D. $[-\pi, \pi]$

102. The function $f(x) = \sin^{-1}(\cos\ x)$ for $x \in [0, \pi]$ has range

A. $[0, \frac{\pi}{2}]$

B. $[-\frac{\pi}{2}, 0]$

C. $[-\frac{\pi}{2}, \frac{\pi}{2}]$

D. $[0, \pi]$

103. The function $g(x) = \cos^{-1}(\sin\ x)$ for $x \in [0, \pi]$ has range

A. $[0, \frac{\pi}{2}]$

B. $[\frac{\pi}{2}, \pi]$

C. $[0, \pi]$

D. $[-\frac{\pi}{2}, \frac{\pi}{2}]$

104. The function $f(x) = \sin^{-1}\ x + \sin^{-1}(1 - x)$ is

A. Increasing on $[0,1]$

B. Decreasing on $[0,1]$

C. Constant on $[0,1]$

D. Neither increasing nor decreasing

105. The function $f(x) = \tan^{-1}(\sqrt{1 - x^2}/x)$ can be rewritten as

A. $\cos^{-1}\ x$ for $0 < x \leq 1$

B. $\sin^{-1}\ x$

C. $\sec^{-1}\ x$

D. $\csc^{-1}\ x$

106. The value of $\tan^{-1}\left(\frac{1-x}{1+x}\right) + \tan^{-1}\ x$ for $-1 < x < 1$ is

A. 0

B. $\frac{\pi}{4}$

C. $\frac{\pi}{2}$

D. π

107. If $0 < x < 1$, then $\sin^{-1}\ x + \sin^{-1}(1 - x)$ lies in

A. $[0, \frac{\pi}{2}]$

B. $[\frac{\pi}{2}, \pi]$

C. $[\frac{\pi}{6}, \frac{5\pi}{6}]$

D. single fixed value

108. The function $f(x) = \tan^{-1} x$ satisfies

A. $f(-x) = -f(x)$

B. $f(-x) = f(x)$

C. $f(-x) = \pi - f(x)$

D. no symmetry

109. The function $g(x) = \sin^{-1} x$ satisfies

A. $g(-x) = -g(x)$

B. $g(-x) = g(x)$

C. $g(-x) = \pi - g(x)$

D. none of these

110. The function $h(x) = \cos^{-1} x$ satisfies

A. $h(-x) = -h(x)$

B. $h(-x) = h(x)$

C. $h(-x) = \pi - h(x)$

D. none of these

SECTION 4: HIGHER NUMERICAL CONCEPTS (Q111–130)

111. If $\theta = \sin^{-1} \frac{3}{5}$, then $\sin 3\theta$ equals

A. $\frac{36}{125}$

B. $\frac{44}{125}$

C. $\frac{117}{125}$

D. $\frac{33}{125}$

112. If $\theta = \tan^{-1} \frac{4}{3}$, then $\cos 3\theta$ equals

A. $\frac{11}{125}$

B. $\frac{44}{125}$

C. $\frac{117}{125}$

D. $\frac{33}{125}$

113. If $\sin^{-1} x + \sin^{-1} y = \frac{\pi}{2}$ and $x, y > 0$, then $\tan^{-1} \frac{x}{y}$ equals

A. $\tan^{-1} \frac{x}{\sqrt{1-x^2}}$

B. $\tan^{-1} \frac{\sqrt{1-y^2}}{y}$

C. $\tan^{-1} \frac{x}{y} = \tan^{-1} \frac{\sqrt{1-y^2}}{y}$

D. $\frac{\pi}{4}$

114. If $x = \sin^{-1} t$, then $\frac{dx}{dt}$ equals

A. $\frac{1}{\sqrt{1-t^2}}$

B. $\sqrt{1-t^2}$

C. $\frac{t}{\sqrt{1-t^2}}$

D. $-\frac{1}{\sqrt{1-t^2}}$

115. If $y = \cos^{-1} t$, then $\frac{dy}{dt}$ equals

A. $\frac{1}{\sqrt{1-t^2}}$

B. $-\frac{1}{\sqrt{1-t^2}}$

C. $\sqrt{1-t^2}$

D. $\frac{t}{\sqrt{1-t^2}}$

116. If $z = \tan^{-1} t$, then $\frac{dz}{dt}$ equals

A. $\frac{1}{1+t^2}$

B. $\frac{1}{\sqrt{1-t^2}}$

C. $\sqrt{1+t^2}$

D. $-\frac{1}{1+t^2}$

117. The derivative of $f(x) = \sin^{-1}(2x\sqrt{1-x^2})$ at $x = \frac{1}{\sqrt{2}}$ is

A. 0

B. 1

C. -1

D. undefined

118. If $f(x) = \cos^{-1}(2x^2 - 1)$, then $f'(x)$ equals

A. $\frac{-4x}{\sqrt{1-(2x^2-1)^2}}$

B. $\frac{4x}{\sqrt{1-(2x^2-1)^2}}$

C. $\frac{-2x}{\sqrt{1-x^2}}$

D. $\frac{2x}{\sqrt{1-x^2}}$

119. The derivative of $y = \tan^{-1}(\sin x)$ is

A. $\frac{\cos x}{1+\sin^2 x}$

B. $\frac{\cos x}{1+\tan^2 x}$

C. $\frac{1}{\sqrt{1-\sin^2 x}}$

D. $\cos x$

120. The derivative of $y = \sin^{-1}(\tan x)$ (where defined) is

A. $\frac{\sec^2 x}{\sqrt{1-\tan^2 x}}$

B. $\frac{\cos x}{\sqrt{1-\sin^2 x}}$

C. $\sec x \tan x$

D. $\frac{\tan x}{\sqrt{1-\tan^2 x}}$

121. If $f(x) = \sin^{-1} x + \cos^{-1} x$, then $f'(x)$ equals

A. 0

B. 1

C. -1

D. undefined

122. The integral $\int \frac{1}{\sqrt{1-x^2}} dx$ equals

A. $\tan^{-1} x + C$

B. $\sin^{-1} x + C$

C. $\cos^{-1} x + C$

D. $\ln |x| + C$

123. The integral $\int \frac{1}{1+x^2} dx$ equals

A. $\sin^{-1} x + C$

B. $\tan^{-1} x + C$

C. $\cos^{-1} x + C$

D. $\ln |x| + C$

124. If $I = \int \frac{1}{\sqrt{1-x^2}} dx$ and $J = \int \frac{1}{1+x^2} dx$, then

A. $I = \sin^{-1} x + C, J = \tan^{-1} x + C$

B. $I = \tan^{-1} x, J = \sin^{-1} x$

C. both equal $\sin^{-1} x$

D. both equal $\tan^{-1} x$

125. The expression $\tan^{-1} \frac{3}{4} + \tan^{-1} \frac{4}{3}$ equals

A. $\frac{\pi}{4}$

B. $\frac{\pi}{2}$

C. π

D. 0

126. If $\theta = \sin^{-1} \frac{3}{5}$, then $\cos 3\theta$ equals

A. $\frac{11}{125}$

B. \dfrac{53}{125}

C. \dfrac{117}{125}

D. \dfrac{36}{125}

127. If $\alpha = \cos^{-1} \frac{3}{5}$, then $\sin 2\alpha$ equals

A. $\frac{24}{25}$

B. $\frac{18}{25}$

C. $\frac{8}{25}$

D. $\frac{6}{25}$

128. The value of $\sin^{-1} \left(\frac{2x\sqrt{1-x^2}}{1-2x^2}\right)$ in terms of $\tan^{-1}$ is most directly related to

A. $\tan^{-1} x$

B. $2\tan^{-1} x$

C. $3\tan^{-1} x$

D. none

129. If $x = \sin^{-1} t$ and $t \in (0,1)$, then $\frac{dx}{d(\cos^{-1} t)}$ equals

A. -1

B. 1

C. 0

D. ∞

130. The number of advanced MCQs in this LEVEL 3 set is closest to

A. 80

B. 100

C. 130

D. 200

⁂

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